Is Silence Golden? Real Effects of Mandatory Disclosure

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Mandatory disclosure provides benefits, but it also entails costs. One cost concerns managerial learning: by discouraging informed trading, disclosure could reduce managers' ability to glean decision-relevant information from prices. Using mandatory segment reporting in the United States, we uncover a reduction in investment-q sensitivity, indicating lower investment efficiency after regulation. Consistent with learning, lower sensitivity is concentrated in firms with more informed trading and lower financing constraints. Constrained firms exhibit no change in investment-q sensitivity, suggesting that they enjoy countervailing benefits via greater financing and stronger governance. Overall, we document a novel link between mandatory disclosure and real effects. (*JEL* G10, G30, M41)

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> When you talk, you are only repeating what you already know. But if you listen, you may learn something new.
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> – Dalai Lama XIV

The economic consequences of mandatory disclosure have been extensively debated (e.g., Stigler 1964; Benston 1973; Fishman and Hagerty 1990; Healy and Palepu 2001; Shleifer and Wolfenzon 2002; Bushee and Leuz 2005; Greenstone et al. 2006). Some studies document benefits to mandating disclosure in securities markets, whereas others are more circumspect and even others advocate repealing some of these statutes (e.g., Easterbrook and Fischel 1984; Leuz and Wysocki 2016; Goldstein and Yang [2017b] for reviews).[1]

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[1] The focus of our study is on mandatory, rather than voluntary, disclosure. It is well known that firms can choose to voluntarily disclose information even if not mandated.

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A common feature of the earlier studies is that they characterize information flows as occurring from the firm to outsiders. In this study, we explore the role of mandatory disclosure from a different angle. Recent theories allow for information to flow in the reverse direction, that is, from the market to the firm. They posit a role for prices in guiding managerial actions when information about the implications of those actions is better known to market participants (collectively) than the manager. This information can be communicated to the manager only via the price formation process (Hayek 1945), thereby creating a price-based feedback to managerial actions (see Bond et al. 2012). We call this the *Learning* perspective.

Bond et al. (2012) introduce a new notion of price informativeness, viz., the extent to which prices *reveal* new information to the manager, which they term "revelatory price efficiency" (RPE). This contrasts with the conventional notion of price informativeness, which refers to how well prices *reflect* future cash flows, and captures the total amount of information in prices. Bond et al. (2012) term this "forecasting price efficiency" (FPE) and note that the value of financial markets for real decisions may not depend on FPE, because some of this information is already known to the decision maker (and incorporated in past decisions). The value of markets, they contend, depends on RPE, which is the amount of new (i.e., previously unknown) information that prices reveal to the decision maker.

The learning perspective introduces a new dimension to the role of disclosure. By impounding information that is *known* to the manager into the price, disclosure could potentially crowd out information that is *unknown* to the manager (and beneficial from a learning perspective). The overall impact of disclosure now becomes ambiguous as the benefits of lower information asymmetry (i.e., more FPE) must be traded off against the costs of lower learning (i.e., less RPE). Gao and Liang (2013) examine such a trade-off by showing that the crowding out effect of disclosure on informed trading has two theoretically opposing effects. On the one hand, disclosure is beneficial because the reduction in adverse selection improves stock liquidity (the FPE channel). On the other hand, the reduction in informed trading can be costly because it reduces the informational feedback effect from prices to managerial actions (the RPE channel).

We study this trade-off in the context of firm investment, a major corporate decision. We follow prior studies (e.g., Bai et al. 2016; Chen et al. 2006; Edmans et al. 2017) and use the traditional investment-q-sensitivity framework (where future investment is regressed on current q) and incorporate disclosure within this framework. The FPE channel predicts an *increase* in investment-q sensitivity when firms disclose, because lower adverse selection allows firms to raise more capital enabling them to better exploit investment opportunities. In addition, disclosure allows stakeholders to better monitor the manager's actions, ensuring that the latter's investment decisions mirror the firm's investment opportunities (e.g., Wurgler 2000).

The RPE channel, on the other hand, provides a more nuanced prediction for investment-q sensitivity, which depends on how the two sources of information (i.e., what the manager knows and what she wishes to learn) interact. If these two sources of information are positively correlated (like in Gao and Liang 2013), then more disclosure about the known factor results in *less* informed trading about both the known and unknown factors. Thus, even if total information in prices (in an FPE sense) increases, the manager places *less* emphasis on the price in guiding her future investment decisions (in an RPE sense), resulting in lower investment-q sensitivity. On the other hand, if the two sources of information are substitutes (like in Goldstein and Yang 2017a), such that more disclosure about the known factor results in *more* informed trading about the unknown factor, then mandatory disclosure could lead to more managerial learning (even in an RPE sense), resulting in greater investment-q sensitivity. Because it is difficult to predict ex ante which theoretical assumption is more valid, we leave this prediction to the data.

We select the mandatory change in segment disclosure enacted under SFAS 131 in the United States as our setting (Section 2 presents a discussion of the pros and cons of this choice). In 1997, after extensive lobbying by analyst groups, the Financial Accounting Standards Board (FASB) issued SFAS 131. The new standard's management approach required that disaggregated segment information be presented to outsiders based on how management internally evaluates the operating performance of its business units. One of the stated objectives was to provide more disaggregated information (e.g., FASB 1996; Berger and Hann 2003). Using a sample of 990 firms that increased the number of segments reported after SFAS 131 and 2,824 firms that did not report more segments because they were already compliant, we verify that mandatory segment reporting did indeed reduce information asymmetry. We find, in a difference-in-differences design, that stock liquidity increased by 13% relative to preregulation levels for affected firms after regulation. These results indicate that mandatory disclosure "leveled the playing field" by decreasing informed traders' information advantage.

Moving to the investment results, we find a more pronounced *decrease* in investment-q sensitivity for affected firms as compared to unaffected firms after regulation. This evidence appears to support the RPE channel, where more disclosure about what the manager knows dissuades informed traders from impounding information that might be useful to the manager. It also suggests that a (presumably unintended) consequence of "leveling the playing field" is to decrease the ability of prices to guide investment, consistent with Gao and Liang (2013). In terms of economic significance, affected firms' reliance on the price to guide future investment fell by 18% more than those for unaffected firms during this period. We find no results using *lagged* investment relative to the year q is measured. We also find no change in the sensitivity of future investment to current cash flows, which is a nonprice measure of investment opportunities. These falsification tests bolster our contention that the effects are due to learning.

We perform additional tests to validate our inferences. First, we differentiate between financially constrained and unconstrained firms, and expect the decrease in investment-q sensitivity after mandatory segment disclosure to be stronger for unconstrained firms. The ability to adjust investments in response to price signals has been shown to be stronger for unconstrained firms (e.g., Bakke and Whited 2010; Chen et al. 2006; Edmans et al. 2017), so the detrimental effect of disclosure is also likely to be more forceful. For constrained firms, more disclosure could provide countervailing benefits by providing information to outsiders such as capital providers. For example, Goldstein and Yang (2017a) examine a learning model in the context of a financially constrained firm, where the decision maker (a capital provider) is different from the one making the disclosure and show that disclosure could provide new information to the decision maker, thus increasing real efficiency.[2]

Using the Hadlock and Pierce (2010) and Whited and Wu (2006) measures of financing constraints, we find that the decrease in investment-q sensitivity is concentrated in affected firms that were unconstrained in the preperiod. These firms experience a 37% decrease in investment-q sensitivity. Constrained firms, on the other hand, experience a statistically insignificant decrease of 3%. We interpret these results as indicating that financially unconstrained firms suffer more severely from mandatory segment disclosure than constrained firms, because the latter may reap some financing-based benefits due to lower adverse selection and cost of capital (e.g., Baker, Stein, and Wurgler 2003) that potentially offset the learning-based costs.

Second, we expect learning-based effects to be stronger for firms with more informed trading because this is where stock prices were presumably providing new information to managers (e.g., Chen et al. 2006). Again, this is what we find: the decrease in investment-q sensitivity for unconstrained firms is around 3 times as large (and statistically different) for firms with more informed trading in the preperiod than for those with less informed trading. Since informed trading increases information asymmetry, the benefits of disclosure could potentially be larger for financially constrained firms—though this is likely to be just one potential channel. We find some weak evidence here. Although constrained firms with more informed trading experience an *increase* in investment-q sensitivity (of 5%) and those with less informed trading experience a decrease, these results are neither statistically significant from zero by themselves nor significantly different from each other.

A decrease in investment-q sensitivity suggests less efficient investing, but it is not a direct indicator. We follow prior studies (e.g., Bai et al. 2016; Chen et al. 2006) and examine the effect of segment reporting on future

[2] They note the absence of such an effect when the information being disclosed is already known to the decision maker.

profitability. Because loss in learning is stronger for firms with more informed trading, we condition the effect on the preperiod level of informed trading. We find that affected firms with more informed trading suffer decreases in future performance.[3] Overall, we interpret these results as evidence that the reduction in learning engendered by mandatory segment disclosure results in lower investment efficiency (in firms with more informed trading in the preperiod).[4]

We conclude our empirical analyses with several sensitivity tests. First, we verify that our inferences are not confounded by the dot-com bubble. Campello and Graham (2013) show that constrained (but not unconstrained) firms exploited soaring valuations by increasing investments, so we might be merely repackaging these results. We verify that this is not the case. Second, we reconcile our inferences with prior studies, such as Cho (2015), that document benefits to mandatory segment disclosure via the governance and agency channels. Finally, we verify that our results are robust to alternative ways of defining firm investment.

Our study offers two main contributions. Ours is the first study in the U. S. context to document the negative real effects of mandatory disclosure via the learning perspective. In doing so, it broadens the economic consequences of mandatory disclosure. Our evidence suggests that disclosure could impose real costs on firms, while providing informational benefits. This is something that has not been demonstrated empirically. We stress that our results should not be interpreted as a blanket recommendation against disclosure regulation, and that one needs to be careful in extrapolating our inferences to other types of disclosure regulations in the United States or to similar legislations globally. The United States is likely to be a special case because not only is its financial markets more sophisticated with respect to price information (potentially increasing learning costs) but it also has strong governance institutions (thus reducing potential governance benefits).

Second, our results caution against attributing a negative connotation to information asymmetry. Most prior studies on the economic consequences of disclosure document decreases in information asymmetry as the basis for ascribing positive attributes to the regulation. Our evidence indicates that informational efficiency and real efficiency need not go hand-in-hand and that the former is neither a necessary nor a sufficient condition for the latter, something long-postulated by economists (e.g., Hirshleifer 1971), but not yet documented in the literature.

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[3] Firms with less informed trading in the preperiod experience an increase in future performance. This result could be consistent with the agency and governance benefits, but it is not robust across all specifications.

[4] In unreported tests, we verify that the decreases in future performance are not due to proprietary costs. First, we control for shifts in market competition and continue to find robust results. Second, we restrict the sample to firms that did not suffer a decrease in market share after regulation and, thus, are less likely to have experienced greater proprietary costs. We continue to find decreases in future performance within this subsample.

1. Hypothesis Development and Experimental Setting

It is well known that in the presence of agency conflicts (à la Jensen and Meckling 1976; Hölmstrom 1979), outsiders ration financing to the firm (e.g., Stiglitz and Weiss 1981). Mandatory disclosure, in this context, is beneficial as it reduces information asymmetry, eases financing constraints and allows the firm to better exploit investment opportunities. This governance/monitoring channel, however, does not explore the institutional features of secondary trading markets. Channels that explore facets of such institutional features provide different (and at times conflicting) implications for how mandatory disclosure influences firm real effects. For example, a mandatory increase in reporting frequency could cause investors to fixate on short-term performance, resulting in myopic investments (e.g., Agarwal et al. 2017; Kraft et al. 2018).

In this study, we explore managerial learning as one such channel. The basic premise is that managers might not be well informed about *all* facets of the firm's operations and might look to prices to glean information to guide their investment decisions. Stock prices are ideal to perform such a role as they impound information possessed by dispersed investors (e.g., Hayek 1945; Grossman and Stiglitz 1980). Stock markets then not only reflect the manager's actions but also guide them. Bond et al. (2012) discuss this dual-role by introducing two facets of stock price informativeness: forecasting price efficiency (FPE), which captures the extent to which stock prices *reflect* all available information, and revelatory price efficiency (RPE), which captures the extent to which prices *reveal* new information to guide managerial decisions. In the presence of this dual role, the implication of mandatory disclosure becomes nuanced. Disclosure could benefit outsiders by improving the extent to which stock prices *reflect* managerial actions (i.e., greater FPE), but it has the potential to hurt the firm by reducing the extent to which stock prices *guide* managerial actions (i.e., less RPE). In the next section, we compare and contrast the conventional and learning models and discuss the role of disclosure within this context.

1.1 Conventional models versus learning models

The point of departure between conventional and learning models is how uncertainty about firm cash flows is modeled. In conventional models (e.g., Diamond 1985; Verrecchia 1982), uncertainty is unidimensional (i.e., emanating from a single random variable), whereas learning models incorporate two dimensions of uncertainty, such as cash flows from assets-in-place versus growth opportunities (e.g., Bai et al. 2016; Edmans et al. 2017; Gao and Liang 2013) or production technology versus market demand (e.g., Goldstein and Yang 2015, 2017a). Further, the manager is assumed to know less than informed traders about one of the factors (typically growth opportunities or market demand), and looks to the price to guide her real decisions, thereby creating the feedback loop from prices to managerial actions. In the context of investment decisions, learning models predict a greater reliance of capital expenditures on the firm's stock price (i.e., a greater investment-q sensitivity) for firms with more informed trading. Bai et al. (2016) incorporate the role of market-based feedback into the classic q-theory of investment, and show that investment decisions depend more on the stock price when the latter incorporates informed traders' private information (about growth opportunities). Chen et al. (2006) provide evidence consistent with learning by showing that firm investments are more sensitive to stock prices when there is more informed trading.

Learning models introduce a new dimension to the role of disclosure. In conventional models, disclosure is generally beneficial (in an FPE sense) as more information about the firm's unidimensional cash flows is incorporated into stock prices via public disclosure rather than through informed trading. Crowding out informed trading lowers information asymmetry and reduces the firm's cost of capital. As a result, the firm is better able to exploit investment opportunities. In addition, more public information allows stakeholders to better monitor the manager's actions. This FPE channel predicts greater investment-q sensitivity.

However, things become nuanced once two-dimensional uncertainty of the *Learning* models is introduced. It is possible that by crowding out informed trading, mandatory disclosure now results in stock prices impounding less information about the factor that managers wish to learn from prices. Gao and Liang (2013) present such a model where firm value depends on two factors: assets-in-place and growth opportunities, which are affected by a common source of uncertainty.[5] Further, the manager cares to learn about how this uncertainty affects growth opportunities rather than assets-in-place. Because informed traders gather information about the common uncertainty and trade on it in the secondary market, the manager can (partially) learn this information from prices, creating a feedback effect from stock prices to investments. The more the informed traders trade on their private information, the more the manager can learn about the value of growth opportunities from the price, and, thus, her investment decisions will be more efficient. However, the greater informed trading comes at a cost. Because investors of the firm experience liquidity shocks, they discount the stock price in the presence of heightened adverse selection brought about by informed trading. Disclosure thus entails both benefits and costs. By publicly revealing the manager's (noisy) information about uncertainty, disclosure benefits the firm by discouraging informed traders' information acquisition, reducing the liquidity discount and increasing the stock price. However, the lower information acquisition entails a cost because the manager now learns less from the price (regarding informed traders' private information about growth opportunities). Thus, even though the price might be more informative in an FPE sense, it could be *less* useful

[5] As noted in their footnote 4, the sources of uncertainty in assets-in-place and growth opportunities need not be the same. The key assumption is that they are positively correlated.

to the manager because it contains less new information in an RPE sense to guide investment. Thus, the overall effect could be a *decrease* in investment-q sensitivity, which we term the "RPE crowding out channel."

It is pertinent to note that mandatory disclosure could also result in *more* learning (even in an RPE sense), if the two sources of information are substitutes (like in Goldstein and Yang 2017a), not complements. For example, if informed traders acquire more information about the unknown factor when the manager discloses more information about the known factor, then mandatory disclosure could potentially engender more learning, resulting in an *increase* in investment-q sensitivity. We term this effect the "RPE crowding in channel."

Given the difficulty in assessing ex ante whether the two information sources are substitutes or complements, we present our main hypothesis in the null form and two alternative hypotheses as follows.[6]

Null hypothesis: Investment-q sensitivity does not change after mandatory disclosure.

Alternative hypothesis 1: Investment-q sensitivity decreases after mandatory disclosure (RPE crowding out channel).

Alternative hypothesis 2: Investment-q sensitivity increases after mandatory disclosure (RPE crowding in channel and/or FPE channel).

1.2 Choice of experimental setting

A persuasive test of the theory requires several ingredients. First, the setting should examine mandatory rather than voluntary disclosure, because the latter entails selection effects that could confound the (causal) effects of disclosure. This still leaves a large set of potential candidates, for example, accounting rule changes mandating disclosures of related party transactions, special purpose entities, or management fringe benefits, to name a few. However, the learning models suggest that the crowding out effect of disclosure is more likely to occur in some situations than others. In particular, mandatory disclosure should have the potential to interfere with informed traders' information acquisition about the specific kind of information that the manager is relatively less informed about and thus more eager to learn.

We select the mandatory change in segment disclosure rules enacted under SFAS 131 in the United States as our setting. This rule required disaggregated segment sales and other information to be presented to outsiders based on how management internally evaluates its business units. One of the stated objectives of SFAS 131 was to provide more disaggregated information (e.g., FASB 1996).

[6] Further, even if disclosure better aligns the manager's and outsiders' information sets (like in the *FPE* channel), it need not lead to an increase in investment-q sensitivity because the disclosed information (which is already known to the manager) might already be reflected in past investment decisions, leaving future investments unaffected by the impounding of this information in the stock price.

On a general level, this setting comports well with the learning framework. For example, Goldstein and Yang (2015, p. 1737) motivate multidimensional uncertainty by noting that it may pertain to "a conglomerate that operates across different industries or business lines."

More specifically, we believe SFAS 131 lends itself to the detection of a potential crowding out effect of mandatory disclosure. As noted in Section 2.1, learning models generally assume two dimensions of uncertainty, viz., assets-in-place (which reflects the firm's existing production technology about which managers have relatively more information) and growth opportunities (which incorporate factors such as market demand and competition about which managers know relatively less and have greater scope to learn). The availability of disaggregated sales information under SFAS 131 is the key here because sales have a natural link to external market demand and the larger competitive environment, information about which the manager may not have and is likely to learn from outsiders (Chen et al. 2006). If informed traders acquire private information about these two sources collectively (like in Gao and Liang 2013), then the manager's disclosure about realized sales generated from assets-in-place can crowd out informed traders' private information acquisition about growth opportunities.

Prior work is suggestive of such an effect at play in the context of SFAS 131. For example, Berger and Hann (2003) find that the some of the new information disclosed under SFAS 131 as compared to the old standard (SFAS 14) was already available to market participants and impounded into prices through private information acquisition. More pertinently, they note that this preavailability of new information is especially so for the firm's revenues than earnings, as the latter tend to be more firm specific. Given that market participants are likely to use their expertise in sector analysis and market conditions to glean information about the firm's (undisclosed) sales, we expect the "joint-information-collection" assumption in Gao and Liang (2013) to hold here. Thus, mandatory disclosure about firm sales is likely to reduce private information about industry/market conditions in prices, thus leading to less learning. Such a mechanism, we contend, is less applicable to other settings (e.g., related party transactions, special purpose entities, and management fringe benefits).

The segment disclosure setting also offers advantages from an empirical standpoint. First,it affected all publicly listed U. S. firms, thus providing a comprehensive sample. The effective year was 1998 for firms with a December year-end and 1999 otherwise. This staggered introduction based on fiscal year-ends mitigates the likelihood that a single market-wide shock might be driving our results (e.g., Greenstone et al. 2006). Second, the regulation was effective in altering firm behavior as evidenced not only by an increase in the number of segments reported after the regulation (Berger and Hann 2003) but also by a reduction in information asymmetry. We validate these findings here. Third, studying segment reporting eschews the need to estimate "reporting quality" proxies that are fraught with estimation errors (see Leuz and Wysocki 2016). In contrast, the number of segments reported can be measured accurately and also has a theoretical link to transparency (e.g., Hayes and Lundholm 1996). Fourth, segment reporting regulation did not affect all firms equally: some firms were already reporting segments as intended by the regulation leaving them unaffected, whereas others had to increase the number of segments they were reporting under the old rule, rendering them affected. Fifth and finally, the time period allows us to exploit ancillary data (such as informed trading) to hone in on learning by rebutting alternative explanations. Such an analysis would not be possible using other settings, such as the increased frequency of financial reporting during the early 1960s and 1970s (e.g., Butler et al. 2007; Fu et al. 2012) or cross-country reporting regulation (e.g., mandatory IFRS adoption by several countries in 2005).[7]

We hasten to add that the segment disclosure setting is no panacea. First, it is difficult to observe whether informed traders impound more or less information about growth opportunities after the firm starts disclosing disaggregated segment sales. As Goldstein and Yang (2015, p. 1738) note "... in the case of a conglomerate, tracking different populations of informed traders might be more challenging, since it is not immediately clear how many investors specialize in one industry versus another." Second, our inferences might not generalize to other settings. This is true for most natural experiments and is especially so here, because disclosure regulations about other factors that the manager might be similarly aware of (say the firm's cost structure) could potentially stimulate *more* informed trading about factors that the manager wishes to learn about, thereby resulting in greater learning.

2. Data and Research Design

This section describes the data sources, the calculation of the variables used in the empirical analysis, and the regression specifications.

2.1 Sample and sources

Our data come from several sources: accounting data from Compustat, stock price and stock liquidity data from CRSP, segment data from the Historical Segments file of Compustat, and probability of informed trading (PIN) data from Brown et al. (2004). To construct the sample, we begin with firms with business segment information (segment type "BUSSEG") on Compustat in the year prior to adoption of SFAS 131 (1997 for a December year-end firm and 1998 for a non-December year-end firm) and in the first year after SFAS

7 Because increased reporting frequency also engenders myopia, which, in turn, could reduce investments, it is difficult to pin down the evidence to learning. IFRS adoption, on the other hand, comingled reporting changes with improvements in institutional features such as enforcement thus reducing the ability to ascribe observed effects to mandatory disclosure.

131 adoption (1998 and 1999, respectively). We exclude firms with missing values for segment sales. This gives us an initial sample of 6,557 firms. The sample reduces to 4,190 firms when we exclude firms with missing firm identifiers (CIK), accounting information, and those whose annual sales differ from aggregated segment sales by more than 1% (this screen follows prior studies, such as Berger and Hann [2003] and Cho [2015]). Further excluding financial services and regulated industries gives us the final sample of 3,814 unique firms.[8] We refer to this sample as the "full sample."

To ensure that our inferences are not confounded by concurrent changes in firm fundamentals (i.e., diversifying mergers and/or spin-offs), we construct a "pure sample" that retains firms without such confounding events. To do so, we follow Berger and Hann (2003, 2007) and use firms' restated segment disclosures under the first 10K reported under SFAS 131. In particular, firms are required to restate their preperiod segment disclosures under the new classification. We exclude firms where the total segment sales as per the restated disclosures differ from that under the old segment disclosures by more than 1%. We apply this screen only to affected firms due to the extensive amount of hand-collection involved in coding the restated segment data.

Appendix A present an example of an affected firm. Ford Motor Company disclosed two business segments (Automotive and Financial Services) in its 1997 10K (the last year under SFAS 14). In discussing its transition to SFAS 131 in 1998, it notes that SFAS 131 resulted in the company now disclosing five segments (two related to the Automobile business and three related to the Financial Services business). Ford increased the number of segments disclosed after the regulation, so we categorize it as an affected firm. Furthermore, we include it in the "pure sample" because its restated 1997 segment disclosures for the two Automobile segments sum to \$123,193 million (\$121,976 million + \$1,217 million), which is within 1% of its original 1997 segment disclosure (\$122,935).[9] Similarly, total Financial Services revenues as per the restated 1997 segment disclosures amount to \$30,692 million (\$17,144 million + \$3,895 million + \$9,653 million), which is what the company reported under the old standard.

In contrast, Appendix B shows Johnson & Johnson as an unaffected firm. It disclosed three segments (Consumer, Pharmaceutical, and Professional) as per its 1997 10K (i.e., under the old reporting standard) and continued to disclose these three segments as per its 1998 10K (i.e., the new standard). The reason is that J&J was reporting segments externally in accordance with its internal reporting rules, and thus was unaffected by the regulation.

Finally, Appendix C presents Sungard Data Systems, an affected firm that is excluded from the "pure sample." The company is classified as an affected

[8] This is similar to the 3,732 firms reported by Berger and Hann (2003) for their full sample tests.

[9] The difference of \$258 million is due to intersegment eliminations that are also reported in the restated 10K.

Table 1
Descriptive statistics

	Full sample		Pure sample	
CHNG_SEG	Firms	%	Firms	%
−3	5	0.13	–	–
−2	19	0.50	3	0.09
−1	144	3.78	33	0.97
0	2,656	69.64	2,656	78.32
1	511	13.40	359	10.59
2	287	7.52	210	6.19
3	129	3.38	91	2.68
4	44	1.15	31	0.91
5	12	0.31	6	0.18
6	6	0.16	2	0.06
8	1	0.03	–	–
	3,814	100.00	3,391	100.00

This panel presents data for the sample of 3,814 firms that were subjected to the mandatory adoption of segment reporting rules under SFAS 131. *CHNG_SEG* denotes the difference between the number of segments that firms disclosed under the new rule (i.e., SFAS 131) in the first year after adoption and the number of segments reported as of the last year under the old rule (i.e., SFAS 14). "Full sample" ("pure sample") denotes firms without any (with) potentially confounding changes in underlying fundamentals such as diversifying acquisitions and/or divestitures.

firm because it self-reported as a single-segment firm in the preperiod (i.e., December 1997) but increased its segments to three (investment support systems, disaster recovery services, and computer services) in the post-period (December 1998). However, total revenues as per the restated segment disclosures for 1997 amount to \$925.03 million (\$629 million + \$228.23 million + \$67.80 million), which is 7% more than the total revenues of \$862.15 million that the company originally reported in December 1997.

Table 1 presents the distribution of changes in the number of segments reported by the sample firms around mandatory disclosure regulation. The change in segments reported varies from a minimum of −3 (i.e., three fewer segments disclosed in the post-period (i.e., SFAS 131) as compared to the preperiod (i.e., SFAS 14) to a maximum of eight. Of the 3,814 firms in the full sample, 2,656 (69.64%) continue to report the same number of segments in the post-period like in the preperiod, whereas 990 firms (25.95%) report more segments. A small proportion of 168 firms (4.41%) report fewer segments.[10] The statistics for the pure sample are similar.

Figure 1 presents a frequency graph of the number of segments reported by the sample firms in each of the periods (pre- and post-regulation). Consistent with prior studies, there is an increase in the number of firms that report multiple segments in the post-period as compared to the preperiod. This indicates that SFAS 131 resulted in an overall increase in the number of segments that firms reported externally. Our final sample comprises 32,324 firm-year observations for the 3,814 firms over the period 1993 to 2004 (i.e., 5 years before and 5 years after the mandatory adoption of SFAS 131).

10 A large majority of these firms (76%) has confounding changes in their underlying fundamentals and thus are excluded from the "pure sample."

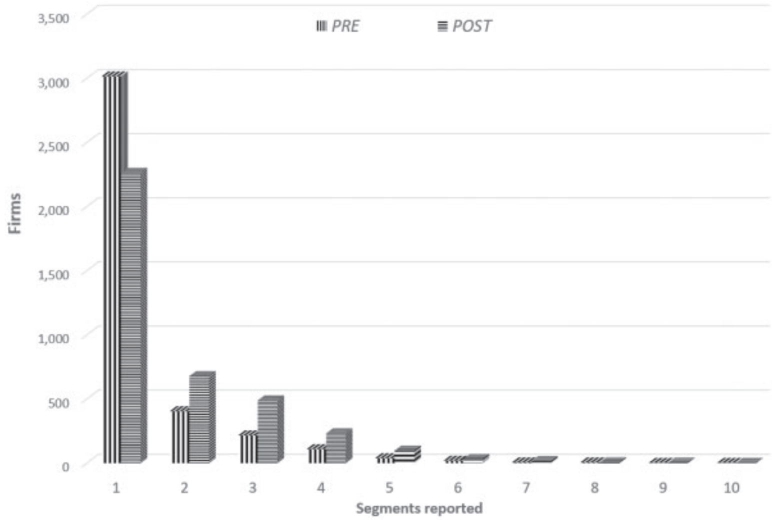

Figure 1
Segments reported by the sample firms
The horizontal axis plots the number of segments reported by the (full) sample firms in each of the periods (i.e., pre- and post-regulation). The vertical axis plots the number of firms reporting these segments. The vertical-lined bar denotes the pre-period, and the horizontal-lined bar denotes the post-period.

2.2 Regression specifications and variable definitions

We test our main hypothesis using a difference-in-differences approach that compares changes in investment-q sensitivity before versus after SFAS 131 adoption for treated (i.e., affected) firms as compared to control (i.e., unaffected) firms. Unlike a conventional difference-in-differences setting, where the regulation affects only some firms, but not others, SFAS 131 affected all U. S. firms around the same time (1998 for a December year-end firm and 1999 for others). To circumvent the lack of a control group, we follow prior studies and classify treated (control) firms as those that disclosed more (same or less) segments under the new regime of SFAS 131 as compared to the old regime of SFAS 14.[11] This design allows us to exploit within-treatment variation in the effect of the regime shift. The disadvantage, however, is that because the regulation was not staggered in event time, the results could be confounded by unobservable factors that also drive investment-q sensitivity. We perform several robustness tests to mitigate this possibility.

Our main specification is a single-stage, firm-level regression where we augment the classical investment-q regression with interactions for SFAS 131,

[11] This classification assigns firms with decreases in the number of segments disclosed as unaffected firms. Although it is possible that reporting fewer segments also reduces information asymmetry, this is not borne out in the data. These firms experience an insignificant change in information asymmetry. Our results are robust to excluding these firms.

as in Equation (1). This design is similar to Edmans et al. (2017) and Foucault and Frésard (2012).

$$
\begin{aligned}
INV_{i,t+1} = {} & \alpha_i + \gamma_t + \beta_1 q_{i,t} + \beta_2 CFO_{i,t} + \beta_3 TREAT*POST_{i,t} + \beta_4 q*TREAT_{i,t} \\
& + \beta_5 q*POST_{i,t} + \beta_6 q*TREAT*POST_{i,t} + \beta_7 CFO*TREAT_{i,t} \\
& + \beta_8 CFO*POST_{i,t} + \beta_9 CFO*TREAT*POST_{i,t} + \beta_{10} SIZE_{i,t} + \varepsilon_{i,t}
\end{aligned} \tag{1}
$$

where, α_i represents firm-fixed effects, γ_t denotes year fixed effects, $INV_{i,t+1}$ represents investment, defined as capital expenditures of firm i as of year $t+1$ scaled by fixed assets as of year t; *TREAT* is a dummy variable that equals 1 for firms with a mandatory increase in transparency (i.e., those disclosing more segments in the post-period as compared to the preperiod) and 0 for all other firms. *POST* is a dummy variable that equals 1 on or after regulation and 0 otherwise. Because the firm and year fixed effects subsume the coefficients on *TREAT* and *POST*, respectively, the latter is omitted from the specification. The former controls for all (time-invariant) heterogeneity across firms, whereas the latter purges overall time trends. We cluster standard errors at the industry level, because these are coarser than firm-level clustering, and especially because mandatory segment disclosure affected several firms in the industry. Our results are, however, robust to clustering by firm.

We use q to represent Tobin's q, defined as the ratio of market value of assets (market value of equity plus book value of debt) divided by book value of assets. q is a price-based measure of a firm's investment opportunity set, and we include *CFO* (cash flow scaled by assets) as a non-price-based measure. This allows us to benchmark the results for q against *CFO* (e.g., Edmans et al. 2017). Finally, we include firm size (*SIZE*) defined as the natural log of market equity in USD millions, following Foucault and Frésard (2012).

The RPE crowding out channel predicts $\beta_6 < 0$ (i.e., a decrease in investment-q sensitivity after mandatory disclosure), whereas the RPE crowding in and FPE channels predict $\beta_6 > 0$.

Panel A of Table 2 provides summary statistics. The average (median) investment rate is 36.9% (22.9%), which is similar to that reported in recent studies (e.g., Foucault and Frésard 2014). The median q is 1.516. Median market equity is \$138 million. Panel B presents preperiod comparisons between affected and unaffected firms. The mean investment rate for affected firms is slightly lower (41.6%) than that for unaffected firms (44.6%), but the median is not statistically different between the two groups. Affected firms are larger in size with an average market value of \$219 million (i.e., $e^{5.393}$), as compared to \$126 million for their unaffected counterparts.

3. Results

3.1 Effect of mandatory segment disclosure on information asymmetry

We begin our empirical analyses by validating that the mandated increase in segment reporting reduced information asymmetry. To do so, we use stock

Table 2
Descriptive statistics

A. Full sample

Variable	Obs.	Mean	Median	SD	Min	Max
TREAT	32,324	0.266	0.000	0.442	0.000	1.000
POST	32,324	0.539	1.000	0.498	0.000	1.000
INV	32,324	0.369	0.229	0.442	0.007	2.819
q	32,324	2.156	1.516	1.852	0.589	11.521
CFO	32,324	0.011	0.075	0.241	−1.244	0.322
SIZE	32,324	5.126	4.930	2.093	1.259	10.621
ILLIQ	32,037	0.685	0.128	1.050	0.000	4.436
PIN	31,523	0.210	0.215	0.106	0.000	0.566
PRC_INV	32,324	0.238	0.094	0.411	0.011	2.703

B. Preperiod comparisons

	Affected firms (N=4,063)		Unaffected firms (N=10,843)	
Variable	Mean	Median	Mean	Median
INV	0.416***	0.268	0.446	0.273
q	2.068***	1.617***	2.321	1.668
CFO	0.056***	0.089***	0.013	0.079
SIZE	5.393***	5.253***	4.842	4.622
ILLIQ	0.570***	0.082***	0.738	0.214
PIN	0.212***	0.214***	0.227	0.230
PRC_INV	0.158***	0.067***	0.208	0.096

Panel A presents descriptive statistics for the entire sample, and Panel B compares preregulation characteristics between affected and unaffected firms. The (full) sample comprises 32,324 firm-year observations for 3,814 unique firms over the period 1993–2004 that corresponds to 5 years before and 5 years after the adoption of mandatory segment reporting. *TREAT* is an indicator variable that takes 1 for firms that disclosed more segments under SFAS 131 as compared to SFAS 141; and 0 for all other firms. *INV* denotes investment and is defined as capital expenditures (data item CAPX) as of year $t+1$ scaled by fixed assets (data item PPENT) as of year t. q denotes Tobin's q as of year t, defined as the market value of equity plus book value of debt (data item AT minus data item CEQ) scaled by book value of assets (data item AT). Market value of equity is defined as shares outstanding (data item CSHO) times closing share price (data item PRCC_F). *CFO* denotes cash flows as of year t, defined as earnings before extraordinary items (data item IB) plus depreciation and amortization (data item DP) scaled by total assets. *SIZE* denotes firm size and is defined as the (log of) market value of equity in USD millions, as of year t. *ILLIQ* denotes (the log of) stock illiquidity. It is defined, following Amihud (2002) as the average of the ratio of daily unsigned stock returns scaled by dollar trading volume multiplied by 10^6. *PIN* denotes the probability of informed trading and measures the ratio of trading by informed traders as a proportion of total trading in the stock. It is obtained from the sources in Brown et al. (2002). *PRC_INV* denotes the inverse of the stock price as of the end of the year. ***, **, and * indicate significance of the difference between affected and unaffected firms at the 1%, 5%, and 10% two-tailed levels.

illiquidity (*ILLIQ*) defined as the Amihud (2002) price impact measure of trades. We present two sets of results: one based on the "full sample" and the other based on the "pure sample." The latter ensures that the results are not due to confounding changes in firm fundamentals.

Panel A of Table 3 presents the results. The main coefficient of interest is that on *TREAT*POST* which captures the incremental effect of mandatory disclosure on stock illiquidity for affected firms as compared to unaffected firms. We present two specifications: Model 1 excludes the firm-level controls (*SIZE* and *PRC_INV*), and Model 2 includes them. The coefficient on *TREAT*POST* is negative and significant in both models (with p-values of .05 and <.01, respectively) indicating an incremental decrease in stock illiquidity (i.e., a decrease in information asymmetry) for affected firms after

Table 3
Effect of mandatory disclosure on information asymmetry

A. Stock illiquidity (ILLIQ)

	ILLIQ			
	Full sample		Pure sample	
Dep. variable	(1)	(2)	(3)	(4)
*TREAT*POST*	−0.065	−0.072	−0.060	−0.070
	[0.026]**	[0.019]***	[0.030]**	[0.020]***
SIZE		−0.292		−0.286
		[0.019]***		[0.019]***
PRC_INV		0.557		0.562
		[0.039]***		[0.042]***
Clustering	Industry	Industry	Industry	Industry
Fixed effects	Firm, year	Firm, year	Firm, year	Firm, year
Adj. R^2	0.685	0.785	0.682	0.783
Obs.	32,037	32,037	28,520	28,520

B. Probability of Informed Trading (PIN)

	PIN			
	Full sample		Pure sample	
Dep. variable	(1)	(2)	(3)	(4)
*TREAT*POST*	−0.006**	−0.006**	−0.005*	−0.006**
	(0.002)	(0.003)	(0.003)	(0.003)
SIZE		−0.026***		−0.026***
		(0.001)		(0.001)
PRC_INV		−0.011***		−0.012***
		(0.002)		(0.003)
Clustering	Industry	Industry	Industry	Industry
Fixed effects	Firm, year	Firm, year	Firm, year	Firm, year
Adj. R^2	0.685	0.785	0.682	0.783
Obs.	32,037	32,037	28,520	28,520

The dependent variable in panel A is *ILLIQ*, defined like in the Amihud (2002) measure of stock illiquidity, and that in panel B is the probability of informed trading (*PIN*). "Full sample" ("pure sample") denotes firms without any (with) potentially confounding changes in underlying fundamentals. *TREAT* is an indicator variable that equals 1 for firms that disclosed more segments under SFAS 131 as compared to SFAS 141 and 0 for all other firms. *POST* denotes that post-SFAS 131 period. It equals 1 for the 5 years after the firm's adoption of SFAS 131 and 0 for the 5 years before. *SIZE* denotes firm size and is defined as the (log of) market value of equity in USD millions. *PRC_INV* denotes the inverse of the stock price as of the end of the year. Robust standard errors, clustered by two-digit SIC industry codes, are in brackets. In addition, all models include firm and year fixed effects that subsume the coefficients on *TREAT* and *POST*, respectively. ***, **, and * indicate significance of the difference between affected and unaffected firms at the 1%, 5%, and 10% two-tailed levels.

the regulation. Models 3 and 4 use the "pure" sample and provide similar evidence that the overall effect of mandatory disclosure on stock illiquidity remains negative and significant.[12] In terms of economic significance based on Model 2, preregulation stock illiquidity of affected firms is 0.569 (untabulated), which translates into a 13% decrease in stock illiquidity (coefficient on *TREAT*POST* of −0.072 divided by 0.569).

To verify the parallel-trends premise that makes the disclosure rule change a legitimate shock for causal inference, we illustrate graphically the sharpness of the mandatory disclosure effect by plotting the illiquidity effect in event time for

[12] Henceforth, we tabulate results only for the "full" sample and footnote those based on the "pure" sample.

affected and unaffected firms. To do so, we orthogonalize *ILLIQ* with respect to firm-level controls and the fixed effects and regress this measure (separately for affected and unaffected firms) on a series of indicator variables each denoting the year relative to the year of segment disclosure (i.e., $-5,-4,\ldots,-1,1,\ldots,5$). The year of passage (i.e., year 0) forms the benchmark and is omitted. Panel A of Figure 2 plots the coefficient estimates (in dots for affected firms and triangles for unaffected firms) together with the 95% confidence interval (solid lines for affected firms and dashed lines for unaffected firms). The graph not only suggests a downward shift in illiquidity for affected firms after mandatory disclosure but also the absence of such an effect for unaffected firms.

To better gauge how the differential effect between affected and unaffected firms behaves around mandatory segment disclosure, Panel B plots the incremental effect for affected firms relative to unaffected firms by event time. The graph shows that the decrease in stock illiquidity for affected firms (relative to control firms) observed in panel A kicks in from the first year after the rule change. In addition, illiquidity does not appear to be trending down in the preperiod, suggesting that the parallel trends assumption is not violated.

Panel B of Table 3 examines the effect of segment disclosure on informed trading using the Probability of Informed Trading (*PIN*) measure of Easley and O'Hara (1992, 2004) and estimated by Brown et al. (2004).[13] Similar to the results for stock illiquidity, the coefficient on *TREAT*POST* is negative and significant in all the models. This is consistent with mandatory disclosure reducing information asymmetry by reducing the informational advantage of informed traders (consistent with Gao and Liang 2013).

3.2 Effect of mandatory segment disclosure on investment sensitivity

Table 4 presents our main results on investment sensitivity. Model 1 has the baseline estimation without the treatment effects, where future investment (i.e., year after q) is regressed on q, our main explanatory variable. We also include cash flow (*CFO*) and control for firm size (*MVE*) to match the specification in Foucault and Frésard (2012, 2014). We standardize q and *CFO* (by deducting their respective sample means and scaling by their respective sample standard deviations) so that the coefficients can be interpreted as the marginal impact of 1 standard deviation. The coefficient of 0.151 (0.088) on q (*CFO*) indicates a 15.1% (8.8%) increase in investment in response to a 1-standard-deviation increase in q (*CFO*). These estimates are similar to those reported in recent studies (e.g., Foucault and Frésard 2014).

Model 2 introduces the effect of the mandatory disclosure. The negative and significant coefficient of -0.029 on *q*TREAT*POST* indicates a decrease in investment-q sensitivity after mandatory segment disclosure. This finding is suggestive of the RPE crowding out channel. This result suggests that managers

13 We thank the authors for making these data available.

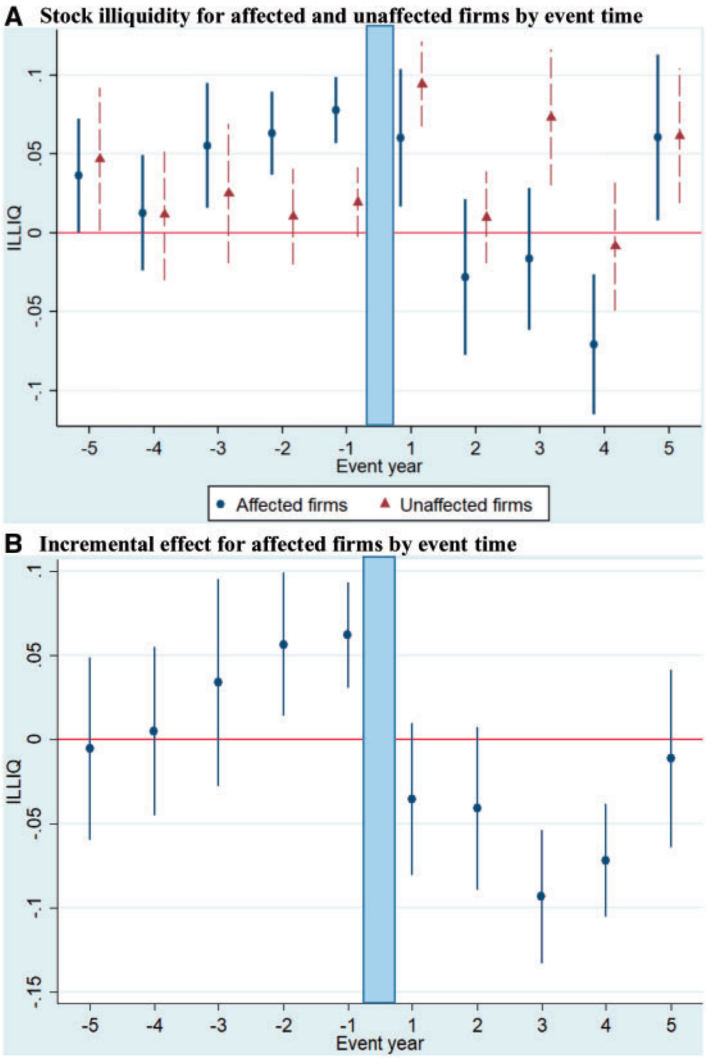

Figure 2
Time trends in information asymmetry (*ILLIQ*)
The x-axis denotes the year relative to the year of regulation. The y-axis plots the stock illiquidity (*ILLIQ*, defined like in the Amihud [2002] measure of stock illiquidity) coefficient for each event-year estimated separately for affected and unaffected firms in panel A. Panel B plots the incremental effect for affected firms estimated using the entire sample. The year of regulation (year 0) is the omitted benchmark. The dots (lines) represent coefficient estimates (95% confidence intervals).

reduce their reliance on stock prices in guiding their investment decisions, presumably due to less revelatory information in prices. A 1-standard-deviation increase in affected firms' q increases investment by 0.161 in the preregulation period (coefficient on q (0.168) + coefficient on q*TREAT* (-0.007). The

Table 4
Effect of mandatory disclosure on investment-*q* sensitivity

	Future investment (INV_{t+1})			Past investment (INV_{t-1})		
Dep. variable	(1)	(2)	(3)	(4)	(5)	(6)
q	0.151	0.168	0.171	−0.006	−0.009	−0.009
	[0.012]***	[0.014]***	[0.014]***	[0.004]	[0.005]*	[0.005]**
CFO	0.088	0.087	0.105	−0.036	−0.036	−0.036
	[0.005]***	[0.005]***	[0.013]***	[0.004]***	[0.004]***	[0.007]***
*TREAT*POST*		−0.010	−0.007		0.003	0.006
		[0.008]	[0.009]		[0.005]	[0.007]
*q*TREAT*		−0.007	−0.008		0.005	0.006
		[0.015]	[0.015]		[0.005]	[0.006]
*q*POST*		−0.025	−0.033		0.004	0.004
		[0.006]***	[0.005]***		[0.005]	[0.005]
*q*TREAT*POST*		−0.029	−0.025		0.003	0.001
		[0.012]**	[0.010]**		[0.005]	[0.006]
*CFO*TREAT*			−0.005			0.015
			[0.022]			[0.009]
*CFO*POST*			−0.026			−0.001
			[0.013]*			[0.006]
*CFO*TREAT*POST*			−0.004			−0.017
			[0.022]			[0.011]
SIZE	−0.004	0.000	0.002	−0.014	−0.015	−0.015
	[0.009]	[0.009]	[0.008]	[0.005]**	[0.006]**	[0.006]**
Clustering	Industry	Industry	Industry	Industry	Industry	Industry
Fixed effects	Firm, year	Firm, year	Firm, year	Firm, year	Firm, year	Firm, year
Adj. R^2	0.300	0.301	0.302	0.36	0.36	0.36
Obs.	32,324	32,324	32,324	30,570	30,570	30,570

This panel presents results for the full sample, where the dependent variable in Models 1 to 3 is future investment (INV_{t+1}), and that in Models 4 to 6 is past investment (INV_{t-1}). All other variables are as of year t. q denotes Tobin's q and has been standardized to have a mean of 0 and a standard deviation of 1. *CFO* denotes cash flows and also has been standardized to have a mean of 0 and a standard deviation of 1. *TREAT* denotes firms that disclosed more segments after mandatory reporting regulation as compared to before. *POST* denotes the post-regulation period. *SIZE* denotes firm size. All regressions include firm and year fixed effects and robust standard errors clustered by two-digit SIC industry (in brackets). ***, **, and * indicate significance of the difference between affected and unaffected firms at the 1%, 5%, and 10% two-tailed levels.

coefficient of −0.029 on *q*TREAT*POST* indicates that this sensitivity reduced by 0.029 or 18% of preperiod levels after mandatory disclosure regulation.[14] Results in Model 3 that introduces the CFO-based interactions indicate that the coefficient on *q*TREAT*POST* remains negative and significant, whereas that on *CFO*TREAT*POST* is insignificant. This suggests that the reliance on cash flows (a nonprice proxy for investment opportunities) remains unchanged after mandatory disclosure, a finding that is expected because the learning perspective is unique to the firm's stock price.[15]

Panel A of Figure 3 presents the graphical evidence for investment-q sensitivity, estimated separately for affected and unaffected firms, with coefficient estimates in dots for affected firms and triangles for unaffected

[14] These estimates capture incremental effects for affected firms. The total decrease in investment-q sensitivity for these firms is 33% ((*q*POST* (−0.025) + *q*TREAT*POST* (−0.029)) / (*q* (0.168) + *q*TREAT* (−0.007)). Unaffected firms experience a decrease in investment-q sensitivity to the tune of 15% (*q*POST* [−0.025] / *q* [0.168]).

[15] The coefficient on *q*TREAT*POST* (*CFO*TREAT*POST*) is significant (insignificant) in the "pure" sample.

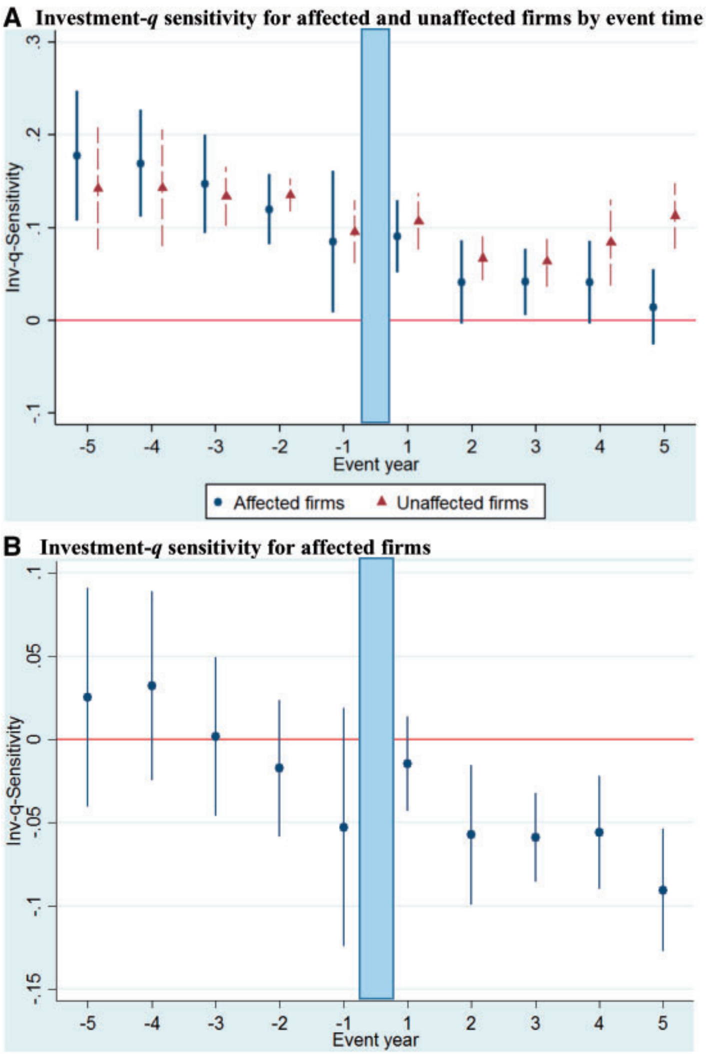

Figure 3
Time trends in investment-*q* sensitivity
The x-axis denotes the year relative to the year of mandatory disclosure regulation, and the y-axis plots the difference-in-differences coefficient estimated for each year relative to the year of mandatory disclosure. Year 0 forms the benchmark and is omitted. The dots represent coefficient estimates and the vertical lines denote 95% confidence intervals.

firms; and confidence intervals in solid lines for affected firms and dashed lines for unaffected firms. Panel B presents the incremental effect for affected firms. Investment-q sensitivity for affected and unaffected firms track each other closely in the preregulation period, suggesting that the "parallel trends" assumption is not violated. Turning to the post-period, there is a decrease in investment-q sensitivity for affected firms, but not for unaffected firms. Panel B presents the differential effect more starkly. The difference-in-differences coefficients seem to bounce around in the preperiod, but none of them are significantly different from 0. In contrast, each of these coefficients (starting from year 2) is statistically less than zero in the post-period. These results not only rule out preperiod trends but also indicate a lag for the reduced information in prices to manifest in investment decisions. This delayed learning effect is also found in other settings (e.g., Edmans et al. [2017] for insider trading enforcement).

Models 4 to 6 of Table 4 present results based on measuring investment as of the year prior to q. This provides a useful falsification, as managers cannot learn from new information in prices that has not yet arrived. Consistent with learning, the coefficient on *q*TREAT*POST* is statistically insignificant and indistinguishable from zero (0.003 and 0.001).

3.3 Cross-sectional tests

While the observed decrease in investment-q sensitivity and the nonresult on investment-CFO sensitivity are consistent with the RPE crowding out channel, these could be driven by other factors that might have changed around the event. We address these alternative explanations by conducting two cross-sectional tests that hone in on the FPE versus RPE trade-off.

3.3.1 Financially constrained versus unconstrained firms. Our first cross-sectional test splits on the (preperiod) level of financing constraints. The theoretical effect of disclosure on firm investment is likely to differ between constrained and unconstrained firms. Because the ability to adjust investments in response to price signals has been shown to be stronger for unconstrained firms (e.g., Bakke and Whited 2010; Chen et al. 2006; Edmans et al.2017), we expect the detrimental impact of disclosure on learning to be larger for these firms. Thus, the decrease in investment-q sensitivity (i.e., RPE crowding out) should be more pronounced for financially unconstrained than financially constrained firms.

There is another channel that differentiates between constrained and unconstrained firms, viz., who is learning from the price. In contrast to Gao and Liang (2013), where the manager makes the disclosure and also learns from the price, Goldstein and Yang (2017a) examine a learning model in the context of a financially constrained firm, where the party learning is a capital provider. They show that in such a scenario, disclosure could provide more new information to the decision maker, thereby reducing capital rationing. They note the absence of such an effect when information being disclosed is already known to the decision maker (i.e., the one making the investment). Thus, it is possible that for constrained firms, disclosure could provide financing-based benefits that might mitigate the learning-based costs.

Because financing constraints are difficult to measure, we use two alternative proxies used in prior work: the Hadlock and Pierce (2010) index and the Whited and Wu (2006) index. We compute these measures for each treatment firm based on preperiod values and categorize firms as constrained versus unconstrained based on the median value within the group of affected firms. We split the *TREAT* indicator into two: *TREAT_UNCONS* and *TREAT_CONS*, where the former denotes affected firms that were financially unconstrained, and the latter indicates affected firms that were constrained. We interact these indicators with *POST* and *q* to assess how the mandatory disclosure effect on investment-*q* sensitivity differs between constrained and unconstrained firms.

Panel A of Table 5 presents the results. We estimate the full specification (Model 3 of Table 5) but tabulate only the relevant variables for parsimony. Models 1 and 2 present results for the HP and WW measures, respectively. The inferences are similar under both these measures, so we discuss the latter only. Consistent with our prediction, the coefficient on *q*TREAT_UNCONS*POST* is more negative (coefficient is −0.077) than that on *q*TREAT_CONS_POST* (coefficient is −0.005), indicating that the detrimental effect of mandatory segment disclosure is more forceful for financially unconstrained firms as compared to constrained firms.[16] Moreover, the coefficient on *q*TREAT_CONS_POST* is insignificant, indicating that the post-disclosure change in investment-*q* sensitivity for financially constrained firms is not statistically different from zero. In terms of economic significance, unconstrained firms experience a 37% decrease in investment-*q* sensitivity, as compared to a (statistically insignificant) decrease of 3% for financially constrained firms. We interpret these results as indicating that financially unconstrained firms suffer more severely from mandatory disclosure. Further, our results suggest that constrained firms might reap other benefits, which offset the learning-based costs, but we caution that this is just one potential explanation.

3.3.2 Preperiod informed trading. Our second cross-sectional test exploits variation in the amount of informed trading in the preperiod. If the decrease in investment-*q* sensitivity is due to less learning, then it should be more pronounced for firms with more informed trading in the preperiod. In other words, the decrease in investment-*q* sensitivity from RPE crowding out that we uncover for financially unconstrained firms should be stronger for those with more informed trading. We also expect differences among financially constrained firms depending on the level of informed trading, because the resultant decrease in information asymmetry should have a larger impact on firms with more preperiod information asymmetry (i.e., more informed trading). To test these predictions, we split the sample based on the median (preperiod)

[16] These coefficients are also statistically different from each other.

Table 5
Cross-sectional tests

A. Financing constraints

	INV_{t+1}	
Dep. variable	HP index [1]	WW index [2]
*q*TREAT_UNCONS*POST* [a]	−0.077	−0.060
	[0.020]***	[0.016]***
*q*TREAT_CONS*POST* [b]	−0.005	−0.012
	[0.014]	[0.012]
p-value of [a] = [b]	.015	.022
Controls	Yes	Yes
Clustering	Industry	Industry
Fixed effects	Firm, year	Firm, year
Adj. R^2	0.302	0.302
Obs.	32,324	32,324

B. Informed trading (PIN)

	INV_{t+1}			
	HP index		WW index	
Dep. variable	Low *PIN* [1]	High *PIN* [2]	Low *PIN* [3]	High *PIN* [4]
*q*TREAT_UNCONS*POST* [a]	−0.049	−0.148	−0.033	−0.109
	[0.032]	[0.037]***	[0.033]	[0.025]***
*q*TREAT_CONS*POST* [b]	−0.013	0.007	−0.024	0.008
	[0.021]	[0.038]	[0.021]	[0.036]
p-value of [a] = [b]	.277	<.01	.794	<.01
p-value of diff. across Low/High split:				
*q*TREAT_UNCONS*POST*	.053		.063	
*q*TREAT_CONS*POST*	.343		.265	
Controls	Yes	Yes	Yes	Yes
Clustering	Industry	Industry	Industry	Industry
Fixed effects	Firm, year	Firm, year	Firm, year	Firm, year
Adj. R^2	0.356	0.259	0.356	0.258
Obs.	16,111	16,102	16,111	16,102

This panel presents results for the full sample, where the dependent variable is future investment (*INV*). *q* denotes Tobin's q and has been standardized to have a mean of 0 and a standard deviation of 1. Panel A splits the *TREAT* indicator into *TREAT_UNCONS* and *TREAT_CONS* depending on whether the firm was financially unconstrained or constrained during the preperiod. Financing constraints are measured using the Hadlock and Pierce (2010) measure in columns titled "HP index" and the Whited and Wu (2006) measure in those titled "WW index." Panel B splits the sample based on the (preregulation) level of *PIN*, where low (high) informed trading indicates below (above) the median. All models include all control variables and fixed effects and are similar to those used in Model 3 of Table 5. Only the relevant coefficients are tabulated for parsimony. Robust standard errors clustered by industry are in brackets. ***, **, and * indicate significance of the difference between affected and unaffected firms at the 1%, 5%, and 10% two-tailed levels. Significance tests of differences across subsamples are one-tailed.

PIN and classify firms below the median as "low *PIN*" firms, whereas those above the median as "high *PIN*." We then estimate our main regression (with the split *TREAT_UNCONS* and *TREAT_CONS* indicators) within each subsample.

Results are presented in Panel B of Table 5. Models 1 and 2 present results based on the HP index, and Models 3 and 4 present those based on the WW index.[17] Consistent across these two measures and in-line with our prediction,

[17] Although we estimate the full specification with all interaction terms and fixed effects, we only present the relevant coefficients for parsimony. The entire table is available on request.

the detrimental effect of disclosure on unconstrained firms is much stronger for those with more (preperiod) informed trading. In particular, the coefficient on *q*TREAT_UNCONS*POST* for high *PIN* firms is −0.148 and almost 3 times as large as that for low *PIN* firms (−0.049). Not only are these coefficients significantly different from one another but also only the former is statistically significant from zero, whereas the latter is not. In other words, only financially unconstrained firms with more informed trading in the preperiod experience a statistically significant decrease in investment-*q* sensitivity after mandatory segment disclosure, whereas those with less informed trading do not.

Turning to financially constrained firms, there is suggestive evidence of a positive but statistically weaker effect of mandatory disclosure. The coefficient on *q*TREAT_CONS*POST* is positive (0.007) for firms with more informed trading, indicating greater investment-*q* sensitivity (to the tune of 5%) after mandatory disclosure. This effect is neither statistically significant from zero nor is it significantly different from that for less informed trading firms (coefficient on *q*TREAT_CONS*POST* is −0.013). Overall, we interpret these results as consistent with learning and the trade-off between higher FPE and lower RPE.

3.4 Additional tests

We perform three additional tests. First, we verify that our inferences are not confounded by the dot-com bubble that occurred during this time. Second, we reconcile our results with Cho (2015) who finds favorable effects of segment disclosure on internal investment efficiency. Third, we ensure that our results are robust to alternative ways of defining investment.

3.4.1 Dot-com bubble. The introduction of SFAS 131 in 1998/1999 coincided with the peak of the dot-com bubble, raising concerns about a confounding effect. For example, if the large increases in valuation during this time were not accompanied by corresponding increases in firm investment, one could potentially observe decreases in investment-*q* sensitivity, but no changes in investment-cash-flow sensitivity. This is precisely what we find. In particular, Campello and Graham's (2013) finding that financially constrained (but not unconstrained) firms exploited their soaring stock prices to increase investment adds suspicion to whether our results are driven by the dot-com bubble. To address these concerns, we exclude firms that were potentially affected by the dot-com bubble and verify whether our inferences hold. These results are presented in Panel A of Table 6. We tabulate only the coefficients on *q*TREAT_UNCONS*POST* and *q*TREAT_CONS*POST*, but estimate the full specification with all interaction terms and fixed effects. Further, we only present results based on the HP measure, but verify that the results are robust to using the WW measure.

We begin in Model 1 by excluding firms in tech industries (SIC codes 481 and 737), and find that our results are robust to excluding these firms. Not only do the coefficients on *q*TREAT_UNCONS*POST* and *q*TREAT_CONS*POST* remain significant and insignificant, respectively, but also they are significantly different from one another. Second, we also exclude firms with manufacturing links to tech industries, which constitute two kinds of firms: (1) technology-related manufacturers, which Campello and Graham (2003) characterize as firms in nonbubble sectors that introduced or demanded technologies related to the innovations of the late 1990s (SIC codes 355, 357, 366, 367, 369, 381, 382, and 384), and (2) firms with supply-chain links to tech industries (SIC codes 227, 229, 348, 351, 361, 362, 365, 371, 372, 376, 379, 385, and 386). Model 2 presents the results and finds that the coefficient on *q*TREAT_UNCONS*POST* remains negative and significant, whereas that on *q*TREAT_CONS*POST* remains insignificant. Model 3 goes one step further and excludes firms with marketing links to tech industries (identified by Campello and Graham [2003] as SIC codes 232, 233, 234, 236, 273, 363, and 394). The inferences again remain unchanged. As a final "catchall" filter, we additionally exclude firms with valuation increases between the start of the bubble and its peak in the top decile.[18] Model 4 continues to uncover a negative and significant coefficient on *q*TREAT_UNCONS*POST* and an insignificant coefficient on *q*TREAT_CONS*POST.* Overall, we interpret these results as suggesting that our results are not driven by the dot-com bubble.

3.4.2 Reconciling with Cho (2015). Our cross-sectional results on constrained versus unconstrained firms suggest how overall inferences about SFAS 131 could differ based on the mix between unconstrained and constrained firms in any sample. In a sample such as ours that uses all firms existing around the passage of SFAS 131, learning-based (i.e., *RPE*) costs appear to dominate any financing/governance (i.e., *FPE*) benefits. However, it is conceivable that a study whose sample is tilted more toward financially constrained firms could conclude that SFAS 131 is beneficial due to an increase in investment-*q* sensitivity for such firms. Cho (2015) is such a study. Using a drastically smaller sample of diversified firms, Cho (2015) uncovers an increase in segment-level investment-*q* sensitivity, which he attributes to SFAS 131 engendering monitoring benefits. It is pertinent to note that Cho (2015) relies on a measure of how internal resources are redistributed among segments depending on their relative growth opportunities, which he notes implicitly assumes that firms are capital-constrained. Cho (2015) points out that if firms are not capital constrained, then all the firm's segments would be allocated additional capital thereby invalidating the approach. After imposing a series of data requirements, Cho's (2015) final sample comprises 1,391 observations that correspond to

[18] In particular, for each firm in the remaining industries, we compute the percentage growth in market-to-book between 1995 (the start of the boom) and 2000 (the end). We then delete firms with percentage increases in the top decile of the entire sample. Our results are robust to excluding the top quintile or the top quartile.

around 348 unique firms, which is only about 10% of our "Full" sample of 3,481 unique firms.

To reconcile with Cho (2015), who examines only diversified firms, we exclude firms that were classified as single-segment in both the pre- and the post-regulation periods and reestimate our main regression on this smaller sample of 13,111 observations. The coefficient on *q*TREAT*POST* remains negative and significant in Model 1 of Panel B, Table 6, suggesting that our results are not driven by the inclusion of single-segment firms. We are unable to approximate Cho's (2015) drastically reduced sample of 348 firms, so we seek to replicate his cross-sectional tests. In particular, Cho (2015) finds that the increase in internal capital market efficiency is stronger for more diversified firms as well as those facing greater takeover threat.

We split our sample of 13,111 observations in two ways: (1) low versus high diversification based on the median post-disclosure segments (as this captures firms' true classifications) and (2) low versus high entrenchment based on the median (state-level) antitakeover measure as of the year prior to regulation (i.e., 1997). We use the state-level measure because few firms in our sample have G-index data. We define *VAR* and set to 1 for firms with high diversification, and alternatively for firms with low managerial entrenchment. To mirror the design of Cho (2015), we interact *VAR* with q, *TREAT* and *POST* to assess how the treatment effect varies depending on the extent of diversification and the strength of governance, respectively.[19]

Results based on setting *VAR* to 1 for high diversification firms are presented in Model 2 of Panel B. Although we estimate the full specification with all the interaction terms and the fixed effects, we tabulate only the coefficients on *q*TREAT*POST* and *q*TREAT*POST*VAR* for parsimony. Whereas the coefficient on *q*TREAT*POST* remains negative and significant, that on *q*TREAT*POST*VAR* is positive (0.034) but insignificant, indicating that affected firms that are highly diversified do not experience a differential change in investment-q sensitivity after SFAS 131. However, these inferences dramatically change when the sample is restricted to financially constrained firms. The coefficient on *q*TREAT*POST*VAR* not only becomes statistically significant (at the 5% level) but also dramatically increases in economic magnitude to 0.137, which is more than 4 times that of the overall sample. This indicates that when focusing on financially constrained firms, SFAS 131 appears to deliver Cho's (2015) result of an increase in investment-q sensitivity for highly diversified firms.

Similar inferences come through in Models 4 and 5, where *VAR* is set to 1 for the less entrenched firms. The coefficient on *q*TREAT*POST* is negative and

[19] The research design in these tests slightly differs from our earlier results, where we split the *TREAT* indicator into *TREAT_CONS* and *TREAT_UNCONS*. We do so to highlight how tilting the composition of the sample toward financially constrained firms could generate Cho's (2015) inferences of an increase in investment-q sensitivity for diversified firms and weakly governed firms post-segment disclosure regulation.

Table 6
Additional analyses

A. Excluding industries and firms potentially affected by the dot-com bubble

	INV_{t+1}			
Dep. variable	Excluding tech industries (1)	Excluding tech industries and those with manufacturing links to tech industries (2)	Excluding tech industries and those with manufacturing and marketing links to tech industries (3)	Excluding tech industries, those with manufacturing and marketing links to tech industries, and "old economy" firms with a price appreciation in the top decile (4)
*q*TREAT_UNCONS*POST* (a)	−0.097	−0.110	−0.112	−0.092
	[0.022]***	[0.030]***	[0.032]***	[0.034]***
*q*TREAT_CONS*POST* (b)	0.002	−0.016	−0.021	−0.013
	[0.017]	[0.032]	[0.032]	[0.036]
p-value of (a) = (b)	.002	.031	.040	.057
Controls	Yes	Yes	Yes	Yes
Clustering	Industry	Industry	Industry	Industry
Fixed effects	Firm, year	Firm, year	Firm, year	Firm, year
Adj. R^2	0.290	0.293	0.293	0.305
Obs.	29,148	21,320	20,635	19,401

B. Reconciling with Cho (2015)

	INV_{t+1}				
		VAR = high diversification		*VAR* = low entrenchment	
Dep. variable	All firms (1)	All firms (2)	Constrained firms (3)	All firms (4)	Constrained firms (5)
*q*TREAT*POST*	−0.040	−0.043	−0.044	−0.054	−0.070
	[0.017]**	[0.017]**	[0.020]**	[0.022]**	[0.023]***
*q*TREAT*POST*VAR*		0.034	0.137	0.020	0.115
		[0.042]	[0.061]**	[0.059]	[0.068]*
Clustering	Industry	Industry	Industry	Industry	Industry
Fixed effects	Firm, year	Firm, year	Firm, year	Firm, year	Firm, year
Adj. R^2	0.306	0.306	0.256	0.314	0.254
Obs.	13,111	13,111	6,552	12,042	6,271

Table 6
Continued

C. Alternative definitions of investment

	Full sample INV_{t+1}	
Dep. variable	Scaling by assets (1)	Including R&D (2)
*q*TREAT_UNCONS*POST*	−0.011	−0.015
	[0.004]***	[0.005]***
*q*TREAT_CONS*POST*	0.004	−0.004
	[0.003]	[0.004]
Controls	Yes	Yes
Clustering	Industry	Industry
Fixed effects	Firm, year	Firm, year
Adj. R^2	0.30	0.524
Obs.	32,324	32,324

Panel A presents results for the full sample, where the dependent variable is future investment (*INV*). q denotes Tobin's q and has been standardized to have a mean of 0 and a standard deviation of 1. The *TREAT* indicator is split into *TREAT_UNCONS* and *TREAT_CONS* depending on whether the firm was financially unconstrained or constrained during the preperiod. Financing constraints are measured using the Hadlock and Pierce (2010) measure. All models include all control variables and fixed effects and are similar to those used in Model 1 of Table 6. Only the relevant coefficients are tabulated for parsimony.

Panel B presents results for multisegment firms; that is, it excludes firms that were single-segment both before and after mandatory segment disclosure. The dependent variable is future investment (*INV*). q denotes Tobin's q and has been standardized to have a mean of zero and a standard deviation of one. *VAR* is an indicator variable that denotes highly diversified firms in Models 2 and 3; and less entrenched firms in Models 4 and 5. High versus low diversification and entrenchment are based on the median values of each variable. All models include control variables and fixed effects and are similar to those used in Model 2 of Table 5. Only the relevant coefficients are tabulated for parsimony.

Panel C presents results for the dependent variable investment (*INV*). All results tabulated pertain to the "full sample." q denotes Tobin's q standardized to have a mean of 0 and a standard deviation of 1. *TREAT_UNCONS* and *TREAT_CONS* split the *TREAT* indicator depending on whether the affected firm is financially unconstrained or constrained. *POST* denotes the post-regulation period. All specifications include the full set of control variables and fixed effects used in Model 3 of Table 5, but the results are not presented for parsimony. All regressions include firm and year fixed effects and robust standard errors clustered by two-digit SIC industry (in brackets).

Robust standard errors clustered by industry are in brackets. ***, **, and * indicate significance of the difference between affected and unaffected firms at the 1%, 5%, and 10% two-tailed levels. Significance tests of differences across subsamples are one-tailed.

significant in both the all-firms sample and the constrained sample. However, the coefficient on *q*TREAT*POST*VAR* is positive but insignificant with a value of 0.02 in the former, but jumps to 0.115 (which is around 6 times greater) and becomes statistically significant (at 10% levels) in the latter.

These results suggest that changes in the mix of constrained and unconstrained firms can alter inferences about the overall implications of SFAS 131.

3.4.3 Alternative ways of defining firm investment. Our final sensitivity test alters the definition of investment in two ways. First, we scale capital expenditures by total assets rather than fixed assets. Model 1 of Panel C, Table 6 shows that the inferences remain unchanged: the coefficient on *q*TREAT_UNCONS*POST* remains negative and significant, whereas that on *q*TREAT_CONS*POST* remains insignificant. Second, we define investment as R&D expenses plus capital expenditures (as compared to just the latter) scaled by total assets. Results in Model 2 of Panel C indicate that our results remain significant.

3.5 Future performance

A decrease in investment-*q* sensitivity suggests less efficient investing, but it is not a direct measure. We follow prior studies (e.g., Chen et al. 2006) and examine future performance to gauge the overall effect on real efficiency. If the firm's own stock price becomes a noisier signal for managerial decision-making, then one would expect a decrease in future performance, because the manager is likely to become more erratic in her investing behavior by potentially underinvesting in profitable businesses and/or overinvesting in less profitable ones.

Table 7 presents results of the regression of future performance (defined like in Chen et al. 2006) as average ROE over the next 3 years (ROE_{t+3}) on *TREAT*POST* and *SIZE*. The coefficient on *TREAT*POST* is insignificant in Model 1 indicating no change in future performance for the full sample of affected firms. Because loss in learning is pronounced for firms with more informed trading, we split *TREAT* into *TREAT_HI_PIN* and *TREAT_LO_PIN* and interact each of these with *POST*. The coefficient on *TREAT_HI_PIN*POST* in Model 2 is negative and significant indicating that affected firms with more informed trading in the preperiod suffer a drop in future performance, a finding consistent with learning.[20,21]

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[20] To ensure that the drop in future performance is not driven by proprietary costs, we perform two additional tests. First, we control for shifts in market competition by including industry-times-year fixed effects. Second, we restrict our sample to firms that did not suffer a decrease in market share between the pre- and post-regulation periods and thus are less likely to have suffered a loss of proprietary information to competitors. In unreported tests, we continue to find a negative and significant coefficient on *TREAT_HI_PIN*POST* in each case.

[21] We are unable to detect differences in future performance between financially constrained and unconstrained firms. This could be because the reduction in the cost of equity (via the FPE channel) might lower the hurdle

Table 7
Future performance

	ROE_{t+3}	
Dep. variable	(1)	(2)
*TREAT*POST*	−0.004	
	[0.022]	
*TREAT_HI_PIN*POST*		−0.090
		[0.046]*
*TREAT_LOW_PIN*POST*		0.056
		[0.024]**
SIZE	−0.037	−0.037
	[0.010]***	[0.010]***
Clustering	Industry	Industry
Fixed effects	Firm, year	Firm, year
Adj. R^2	0.30	0.30
Obs.	28,994	28,994

This panel presents results for the full sample, where the dependent variable is average return on equity over the subsequent 3 years (ROE_{t+3}). *TREAT* is an indicator variable that equals 1 for firms that disclosed more segments after segment disclosure regulation and 0 for all other firms. *POST* denotes the post-regulation period. *TREAT_HI_PIN* and *TREAT_LOW_PIN* split the *TREAT* indicator based on the (preregulation) level of *PIN*, where low (high) indicates below (above) the median. *SIZE* denotes firm size. All models include firm and year fixed effects and robust standard errors, clustered by industry. ***, **, and * indicate significance of the difference between affected and unaffected firms at the 1%, 5%, and 10% two-tailed levels.

4. Conclusion

Mandatory disclosure has the potential to impose real costs on firms because it can crowd out investor private information acquisition and interfere with managerial learning. This can lead to a reduced sensitivity of investment to q because of less revelatory information in prices. Our study is the first to empirically document this learning-related cost of mandatory disclosure. Exploiting the setting of a mandatory change in segment disclosure regulation, we find reduced investment-q sensitivity at firms that increased segment disclosure, consistent with a crowding out effect. This decrease in investment-q sensitivity is concentrated in financially unconstrained firms and those with more preperiod informed trading.

We contribute to the disclosure literature by documenting a managerial learning-related cost that has not been empirically demonstrated, thus expanding the set of economic consequences attributed to mandatory disclosure. We hasten to add that our findings do not contradict the idea that mandatory disclosure brings important benefits, which in our setting manifest for financially constrained firms. We also recognize that there are other costs of disclosure such as proprietary costs, although these are unlikely to explain our findings.

rate for new projects, resulting in lower accounting performance. Because the cost of equity is not reflected as an expense on the income statement, the reduction in funding costs is unlikely to flow into the accounting performance measure.

Appendix A. Ford Motor Co. (Affected Firm and Pure)

A.1 Preregulation Disclosure (Two Segments)

Nature of Operations
- --------------------

The company operates in two principal business segments. The Automotive segment consists of the design, manufacture, assembly and sale of cars, trucks and related parts and accessories. The Financial Services segment consists primarily of financing operations, vehicle and equipment leasing and rental operations, and insurance operations.

Automotive
- ----------

	1997
<S>	<C>
Sales to unaffiliated customers	
United States	$ 81,313
Europe	24,424
All other	17,198
Total	$122,935

Financial Services
- ------------------

	1997
<S>	<C>
Revenues	
United States	$ 24,268
Europe	3,194
All other	3,230
Total	$ 30,692

A.2 Post-regulation Disclosure (Five Segments)

NOTE 17. Segment Information
- -----------------------------

Ford adopted Statement of Financial Accounting Standards No. 131, Disclosures about Segments of an Enterprise and Related Information, effective with year-end 1998. This standard requires companies to disclose selected financial data by operating segment (defined in Note 1). Ford has identified four primary operating segments: Automotive, Visteon, Ford Credit, and Hertz. Segment selection was based upon internal organizational structure, the way in which these operations are managed and their performance evaluated by management and Ford's Board of Directors, the availability of separate financial results, and materiality considerations. Segment detail is summarized as follows (in millions):

	Auto-motive	Visteon	Ford Credit	Hertz	Other Fin Svcs
<S>	<C>	<C>	<C>	<C>	<C>
1998					
Revenues					
External customer revenues	$118,017	$ 1,412 a/	$ 19,095	$ 4,241	$ 1,997
1997					
Revenues					
External customer revenues	$121,976	$ 1,217 a/	$ 17,144	$ 3,895	$ 9,653

Appendix B. Johnson & Johnson (Unaffected Firm)

B.1 Preregulation Disclosure (Three Segments)

SEGMENTS OF BUSINESS; GEOGRAPHIC AREAS

Johnson & Johnson's worldwide business is divided into three segments: Consumer, Pharmaceutical and Professional. Johnson & Johnson further categorizes its sales and operating profit by major geographic areas of the world. The narrative and tabular (but not the graphic) descriptions of segments and geographic categories captioned "Management's Discussion and Analysis of Results of Operations and Financial Condition -- Segments of Business, Consumer, Pharmaceutical, Professional and Geographic Areas" on pages 29 through 31 and 44 of Johnson & Johnson's Annual Report to Shareowners for fiscal year 1997 are incorporated herein by reference thereto.

Sales			Increase	
(Millions of Dollars)	1997	1996	Amount	Percent
<S>	<C>	<C>	<C>	<C>
Consumer	$ 6,498	6,364	134	2.1%
Pharmaceutical	7,696	7,188	508	7.1
Professional	8,435	8,068	367	4.5
Worldwide total	$22,629	21,620	1,009	4.7%

B.2 Post-regulation Disclosure (Three Segments)

SEGMENTS OF BUSINESS; GEOGRAPHIC AREAS

Johnson & Johnson's worldwide business is divided into three segments: Consumer, Pharmaceutical and Professional. Johnson & Johnson further categorizes its sales and operating profit by major geographic areas of the world. Additional information required by this item is incorporated herein by reference to the narrative and tabular (but not the graphic) descriptions of segments and geographic areas captioned "Management's Discussion and Analysis of Results of Operations and Financial Condition -- Segments of Business, Consumer, Pharmaceutical, Professional and Geographic Areas" on pages 27 through 30 and 45 of Johnson & Johnson's Annual Report to Shareowners for fiscal year 1998.

Sales			Increase	
(Millions of Dollars)	1998	1997	Amount	Percent
<S>	<C>	<C>	<C>	<C>
Consumer	$ 6,526	6,498	28	0.4%
Pharmaceutical	8,562	7,696	866	11.3
Professional	8,569	8,435	134	1.6
Worldwide total	$23,657	22,629	1,028	4.5%

Appendix C. Sungard Data Systems (Affected Firm, but Not Pure)

C.1 Preregulation Disclosures (One Segment)

SunGard Data Systems Inc. (the Company), through its wholly owned subsidiaries, operates in a single industry segment, principally in the United States, providing computer services, principally proprietary processing services and software to the financial services industry, computer disaster recovery services and healthcare information systems.

(IN THOUSANDS, EXCEPT PER-SHARE AMOUNTS)	YEAR ENDED DECEMBER 31, 1997	1996	1995
<S>	<C>	<C>	<C>
REVENUES	$ 862,151	$ 670,309	$ 532,628

C.2 Post-regulation Disclosure (Three Segments)

For accounting purposes, the Company has three operating segments: investment support systems (ISS), disaster recovery services (DRS), and computer services and other (CS and other). The Company's operating segments are groups of businesses that offer similar products and services. They are managed separately because each business requires different technology and marketing strategies.

1998	ISS	DRS	CS and other	total operating segments
<S>	<C>	<C>	<C>	<C>
Revenues	$ 804,875	$ 275,282	$ 79,591	$1,159,748
Depreciation and amortization	60,970	40,779	5,663	107,412
Operating income	151,191	64,854	12,506	228,551
Total assets	737,829	175,549	34,937	948,315
Cash paid for property and equipment	18,962	37,935	4,605	61,502

<CAPTION>

1997	ISS	DRS	CS and other	total operating segments
<S>	<C>	<C>	<C>	<C>
Revenues	$ 628,992	$ 228,259	$ 67,779	$ 925,030
Depreciation and amortization	56,688	34,704	4,767	96,159
Operating income	110,263	49,183	10,351	169,797
Total assets	632,796	152,099	29,011	813,906
Cash paid for property and equipment	16,728	35,213	4,406	56,347

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